From Clerks to Code: How Technology Impacted Labor in Asset Management?

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Abstract

Financial firms have gone through three major technological waves: computerization in the 1980s and 1990s, the rise of indexing and passive investing in the 2000s and 2010s, and the AI and automation wave from roughly 2015 to the present. This project studies how much labor is required to manage capital across those waves by tracking a simple productivity measure: assets under management per employee. Using a small panel of representative firms, we compare changes in AUM per employee, revenue per employee, and operating expense intensity over time. The goal is not to identify causal effects, but to document stylized facts about how technology changes the scale of asset management work.

1 Introduction

Asset management has changed dramatically over the last four decades. In the earlier computerization era, firms invested in information technology, electronic trading, and back-office automation. Later, the spread of index funds and exchange-traded funds made it possible to run much larger pools of capital with comparatively fewer workers. In the most recent AI era, firms increasingly use machine learning, data platforms, and automation to support research, distribution, compliance, and operations.

This project asks a simple question: how much labor is needed to manage capital as financial technology evolves? The central metric is assets under management per employee, which we treat as a proxy for labor productivity in asset management. A rising ratio suggests that each worker is supporting more capital, whether because of automation, standardization, scale economies, or changes in product mix.

2 Core Idea

The empirical design is intentionally simple and transparent. We construct firm-year observations for a small set of large asset managers and measure how their labor productivity evolves across three broad technology phases:

- Computerization, approximately 1985–2000
- Indexing, approximately 2000–2015
- AI and automation, approximately 2015–present

The core hypothesis is that each successive wave of technology increases the amount of capital that can be managed per employee. The project does not claim a clean causal estimate. Instead, it documents a stylized relationship between technology adoption and labor efficiency.

3 Firms

We keep the sample small but representative by focusing on four large firms with different business models and long operating histories:

Table 1: Representative firms in the sample

Firm	Model	Why it is useful
J.P. Morgan	Scale, AI, ETFs, broad financial services	Captures large-platform adoption of technology
Vanguard Group	Pure passive pioneer	Highlights low-labor passive management model
Fidelity Investments	Active management and retail orientation	Provides contrast to passive-heavy firms
State Street Global Advisors	ETF and institutional focus	Useful for passive and institutional scale comparisons

This selection covers active versus passive management, different adoption speeds, and enough time depth to observe broad trends.

4 Periodization

We divide the sample into three historical phases:

4.1 Computerization, 1985–2000

This period covers the rise of IT infrastructure, electronic trading, and back-office automation. The expected pattern is gradual productivity improvement, especially through the reduction of clerical work and manual processing.

4.2 Indexing, 2000–2015

This period covers the expansion of index funds and ETFs. The expected pattern is a sharper rise in AUM per employee because passive products can scale with fewer operational and research resources.

4.3 AI and Automation, 2015–present

This period covers AI, big data, and quant platforms. The expected pattern is continued growth in productivity, though the slope may flatten as firms shift labor composition rather than simply reduce headcount.

5 Key Metrics

For each firm-year observation, we keep the measurement set minimal.

5.1 Main metric

The main outcome is

$$\text{AUM per Employee} = \frac{\text{Assets Under Management}}{\text{Employees}}.$$

We interpret this as labor productivity in asset management.

5.2 Supporting metrics

We also track:

- Revenue per employee
- Operating expense divided by AUM
- Passive share of AUM, when available or approximable

These secondary measures help distinguish scale effects from margin effects and provide a broader view of organizational efficiency.

6 Data Pipeline

The project is designed to be feasible with a mix of public data and light manual collection.

6.1 Firm financials

The core firm-level financial variables can come from Compustat via WRDS, including employees, revenue, operating expenses, and firm identifiers such as GVKEY. This works especially well for public firms such as J.P. Morgan and State Street Corporation.

6.2 AUM data

The most important variable is AUM. For public firms, this can often be collected from annual reports and 10-K filings. For firms with less standardized disclosure, such as Vanguard and Fidelity, annual reports, fact sheets, and company websites can be used to build a panel manually.

Table 2: Target dataset structure

Firm	Year	AUM	Employees	Revenue	AUM/Employee	Phase
J.P. Morgan	1995	—	—	—	—	Computerization
Vanguard	2005	—	—	—	—	Indexing
Fidelity	2018	—	—	—	—	AI / Automation
State Street	2022	—	—	—	—	AI / Automation

6.3 Passive exposure

If data access allows, passive share of AUM can be estimated from Morningstar Direct or CRSP Mutual Fund data. A rough ETF share of total AUM is still useful for a first pass.

6.4 AI exposure

AI exposure is treated lightly. A simple approach is to count mentions of "AI" or "machine learning" in annual reports, or to include a post-2015 dummy variable as a coarse indicator of the AI era.

6.5 Final dataset

The final panel can be assembled in pandas by merging firm financials with manual AUM data and then computing productivity ratios:

```
df["aum_per_employee"] = df["AUM"] / df["EMP"]
```

7 Output

The paper will focus on three main figures.

7.1 Figure 1

Time series of AUM per employee by firm. This is the main figure and should carry the main narrative.

7.2 Figure 2

Growth rate of AUM versus growth rate of employees. This figure helps show whether capital scale is rising faster than labor input.

7.3 Figure 3

Passive share versus labor productivity. This optional figure can help connect product mix to efficiency.

Figure 1: Placeholder for AUM per employee time series by firm.

8 Narrative Structure

The discussion section will follow the three historical phases.

8.1 Computerization

The expected finding is slow but visible improvement in productivity, mostly through the reduction of clerical labor and improved operating systems.

8.2 Indexing

The expected finding is a sharper increase in AUM per employee. Passive funds require fewer workers per dollar of capital because they are simpler to run at scale.

8.3 AI and Automation

The expected finding is continued growth in productivity, possibly with some flattening if firms shift toward higher-skilled labor rather than pure headcount reduction.

9 Why This Design Works

The design works because it stays close to what the data can support.

- It does not require a strong causal identification strategy.
- It focuses on stylized facts that are easy to communicate.
- It directly links technology, scalability, and labor efficiency.

10 Expected Findings

The likely pattern is:

- Computerization: gradual productivity improvement
- Indexing: large jump in AUM per employee
- AI era: continued growth, potentially at a slower pace

An additional comparison between passive-heavy and active-heavy firms may show a divergence similar to Vanguard versus Fidelity.

11 Conclusion

This project provides a simple empirical framework for studying how technology changes labor demand in asset management. By using AUM per employee as the central metric and organizing the sample into broad technological eras, the paper can present a clear and intuitive account of how firms scale capital with fewer workers over time.